Frequency-Multiplexed Closed-Loop Control of a Brain–Machine Hand Interface

GG4 — Neural Data Analysis with Adaptive State Estimation
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Abstract

I drive a simulated hand to Cartesian targets by commanding a two-dimensional stimulation of a stochastic "brain" whose noisy 16-D activity passes through a fixed, stateful, nonlinear decoder before moving a two-link arm. The end-to-end map is nonlinear, partially observable, non-invertible and noisy, so no standard linear controller applies directly. The contribution is a control *abstraction* obtained by reverse-engineering the decoder: it is a *frequency-addressed muscle bank*, and a small sinusoidal "select tone" at a muscle's frequency, exploiting the fact that the head's power channel is held on "for free" by a latent DC offset, drives that muscle. This reduces control to two independent inverse-kinematics joint servos closed on the hand position. The controller reaches the upper, well-actuated half of the workspace to within ∼6 cm and generalises across brains (median 10 cm, 93% within 15 cm), but the brain's low-pass character makes the high-frequency (elbow) muscles weak, producing a sharp, directionally-structured failure in the lower workspace. I quantify this, contrast a hand-feedback (V1) and a neural-observer (V2) variant to measure the value of sensing (∼24 cm), and show the binding precision limit is a property of the plant, not the control law.

1 Introduction

The GG4 platform poses two coupled problems: *estimation* — recovering hidden structure from noisy neural observations — and *control* — commanding inputs to produce a desired behaviour. This report concerns the final control task: steering a hand to a target (x, y) using only the official interface, namely the per-step neural measurement `measure()`, the current hand position `hand_pos`, and the input `next_state(u)` with $u \in [0, 1]^2$.

The plant is a composition:

$$u \to \underbrace{\text{brain}}_{\text{LGSSM, random}} \to \text{neural} \to \underbrace{\text{decoder}}_{(x_1,x_2)} \to \underbrace{\text{head}}_{\text{nonlinear}} \to 4 \text{ muscles} \to \text{arm} \to \text{hand.}$$

It is nonlinear, partially observable, non-invertible and stochastic, and the hard part is not any single stage but the *interface* between a linear, identifiable front end (the brain) and a nonlinear, rhythm-coded back end. The first half of the work was therefore reverse-engineering the plant until a control abstraction emerged; the second half was a controller built on it and its honest evaluation. The full design log is given in the accompanying lab notebook.

2 Plant structure and key findings

I treat the per-trial *brain* as unknown but identifiable, and the *decoder + head + arm* as a *fixed, brain-invariant* apparatus, identical on every trial. This split is the central structural lever: the random part of the plant is the easy linear part, and the hard nonlinear part never changes. The following were established empirically; several required an exact NumPy replica of the head, verified to match the supplied Torch network to 3×10^{-7} and used as a glass-box to read its hidden internal state.

Frequency-addressed selection. The decoder is linear, $(x_1, x_2) = D\,y + b$. A bank of cosine/sine band-pass filters acts on the recent history of x_1; each of the four muscles responds to a distinct frequency ($\approx 0.073, 0.133, 0.207, 0.312$ cycles/sample). Driving x_1 to oscillate near a muscle's frequency selects it, and the selector saturates easily, so *selection is the easy half.*

Power is "free". The head enables a muscle only when x_2 clears a wake threshold ($\approx +0.72$) with persistent evidence. Crucially, x_2 sits at a large positive DC offset ($\approx +1.8$) under neutral input that already exceeds threshold, and the head's evidence rule (the "same-sign-a-period-apart" weight 0.30 exceeds the evidence threshold 0.16) means a *constant-high* x_2 sustains power indefinitely: feeding the replica a constant $x_2 = 1.5$ holds power at 0.654 ± 0.00. We therefore never synthesise a power signal; we only avoid letting x_2 dip below threshold.

Velocity, not position. Muscle activation is power $\times$ selector $\in [0, 1]$; the arm converts antagonist differences into joint angular *velocity* through dry-(stiction) and wet-(viscous) friction. Hand position is thus the time-integral of a velocity we modulate, and below the stiction threshold the joint *holds* — a free brake.

Low-pass, noisy brain. Because the brain is an approximately LTI state-space system, a tone at frequency f in u produces latent oscillation at the same f, with amplitude and phase set by the brain's transfer function $G(f)$. System identification (§3) shows G is low-pass: the spectral radius is $\rho(A) \approx 0.9$–0.95 and the input–output coherence collapses above ~ 0.1 cyc/sample. Consequently the higher-frequency muscles receive less drive and are weaker — the single fact behind the controller's dominant failure mode. Finally, a single `measure()` is very noisy ($\sigma \approx 2.6$); the arm is driven by a 100-sample average ($\sigma \approx 0.25$), but the controller sees only the single sample, so the loop must be closed and evaluation must be multi-trial.

3 Estimation: brain identification and the hand observer

Estimation enters the control pipeline in three places, reusing the linear-Gaussian state-space machinery developed earlier in the project.

Brain system identification. Driving the brain with a persistently-exciting bounded probe and fitting an LGSSM by expectation–maximisation (with a subspace warm start) yields (A, B, C, Q, R). The identified model's frequency response, compared against an empirical transfer function measured by a periodic multisine (Fig. 1), characterises the brain: its dominant principal gain falls roughly an order of magnitude across the band, so it is open-loop stable and low-pass — exactly what motivates and explains the muscle-strength asymmetry of §2. Identification is well posed here because, unlike the blind setting of the earlier estimation weeks, the inputs are known.

Latent decoding. Because the decoder D is fixed and known, the control-relevant latents $(x_1, x_2) = Dy + b$ are computable from the neural observations even though the head exposes only muscle outputs. This is what allows the glass-box analysis and the input-design reasoning.

Hand-state estimation. The controller needs the joint state. In V1 this is read directly from the measured hand position by inverse kinematics. In V2 it is produced by a forward observer that replays the fixed downstream (decoder$\to$head$\to$arm-velocity) on the observed neural and integrates a joint-angle estimate, using no hand feedback. The two variants are precisely a Kalman *predict-plus-update*

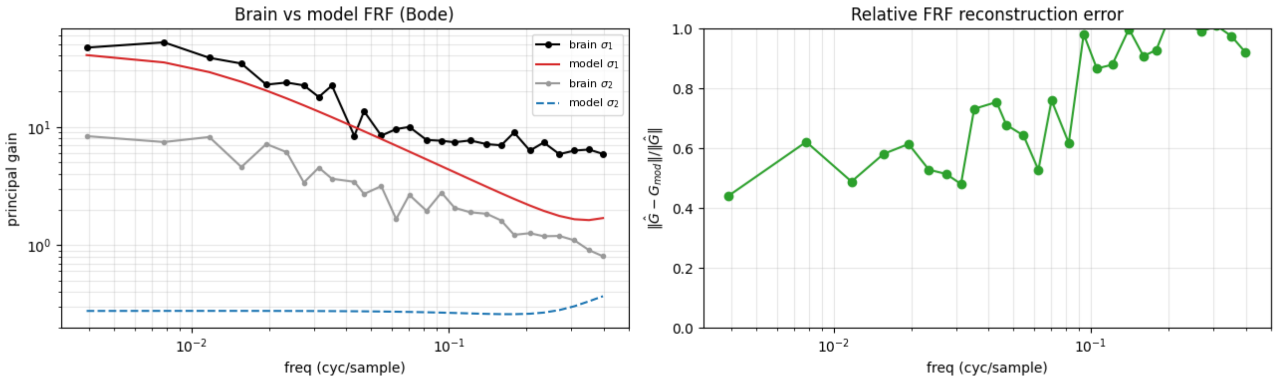

Figure 1: Brain frequency response. Left: measured (black/grey) vs. identified-model (red/blue) principal gains. The dominant gain σ_1 rolls off from $\sim$40 to $\sim$6 with frequency — a direct measurement of the low-pass character that makes the high-frequency (elbow) muscles weak. Right: the model's relative reconstruction error grows with frequency, where the brain transmits little and coherence is lost.

versus a *predict-only* estimator applied to the hand, and their gap (§6) measures the value of the hand measurement.

4 Control approach

4.1 Why standard linear control does not transfer

LQG regulates a linear state to a *static* setpoint, but the hand is not in the brain's state (there is no arm$\to$brain feedback) and the head needs *oscillation* to move at all — the opposite of settling to a constant. Feed-forward Fourier/SVD inversion of the brain can synthesise any reachable *latent* trajectory, but cannot cross the nonlinearity to a position reference and has no mechanism to correct the integrator under noise. Nonlinear MPC could in principle target the hand but requires a full model of the unknown, stochastic stack and a non-convex solve. The conclusion is structural: linear tools belong on the inner channel (produce the right latent oscillation), and the outer problem (position, through a nonlinear integrator) must be closed with feedback on an estimated velocity.

4.2 Design ladder: how the controller was found

The final controller is the end of a short ladder of designs, each discarded for a specific, observable reason — a process worth recording because it justifies the final form rather than asserting it. *(i) Naive two-tone.* Following the literal description of the head, drive x_1 with a select tone and x_2 with a slow power tone. This fails: the select tone, transported into both latents, contaminates x_2, which dips through the wake threshold; because the head's power exits fast but re-enters slowly, each dip crashes power and it recovers only slowly, giving a single "fire-once" slew followed by an un-retriggerable latch-off. *(ii) Clean separation by input-direction nulling.* The principled fix uses the two channels to drive x_1 while *nulling* x_2 at the select frequency — a well-conditioned 2×2 solve. The selection nulling works, but the full scheme also requires a clean, centred x_2 power oscillation, and centring the large $+1.8$ offset consumes nearly all of the $[0, 1]$ headroom, so the select tone clips and clipping re-introduces the contamination; moreover, measuring the per-frequency response online costs $\sim$1500 steps, over the 400-step budget. *(iii) Power from the offset.* The breakthrough (§2) dissolves the problem: x_2's natural offset already holds power on, so the power tone and the centring are dropped entirely, and the only remaining job — a small select tone — is so small that x_2 never dips, making the nulling unnecessary too. The result is the simple, affordable, single-direction drive used below.

4.3 The select-tone joint servo (V1)

The design exploits the structure of §2 directly.

Inner channel. To fire a muscle I drive a sinusoidal select tone at its frequency; power comes free from the x_2 offset, so no power tone and no DC-centring are needed. The tone amplitude is kept small enough that its inevitable leakage into x_2 never pulls it below the $+0.72$ threshold, and that the input never clips $[0, 1]$, while remaining large enough that x_1 clears the 0.15 selector threshold.

Decoupling. The shoulder is driven by muscles 0/1 and the elbow by 2/3, and each joint's velocity depends only on its own pair, so control decouples into two independent one-dimensional joint servos. Because the muscle→joint map is brain-invariant and the arm geometry (link lengths) is known — it is the same geometry the evaluation harness assumes, and is *not* the neural-network weights — I control in inverse-kinematics joint space.

Outer loop. Each step, the target and current hand positions are inverted to joint angles; for each joint whose error exceeds a deadzone, a select tone is emitted with amplitude $a = \mathrm{clip}(k\,|e|,\, a_{\min}, a_{\max})$ (the elbow scaled by a boost, and a total cap to share the input budget). Inside the deadzone the tone is dropped and stiction holds the pose. The frequencies and joint map are calibrated once offline (Fig. 2); the per-brain gain that varies between trials is absorbed by the loop rather than re-identified.

4.4 The neural-observer variant (V2)

V2 keeps the entire servo and replaces only the joint estimate with the forward observer of §3, using no hand feedback. Its difficulty is noise: the observer is fed the single noisy `measure()` whereas the arm responds to the 100-sample average, so the nonlinear head amplifies the $\sim 10\times$ noise mismatch into accumulating drift. The V1–V2 comparison is thus a controlled measurement of how much direct hand sensing is worth.

5 Experimental design and evaluation methodology

All controllers are evaluated through the group-standard harness so that numbers are commensurable: an identical polar grid of targets (radii $\{20, 38, 55\}$ cm $\times$ 8 angles), a fixed brain seed, a fixed horizon ($T = 1000$), and the same six metrics. The grid spans the full reachable workspace rather than a few hand-picked points, which is essential here because the failure is *directionally structured* and a few targets would hide it. The six metrics are chosen to expose distinct aspects: *final distance* (the headline task error), *time-to-threshold* (responsiveness), *steady-state distance* (settling/precision), *shoulder and elbow angle error* (a diagnostic decomposition that localises the failure), *control effort* (efficiency, and a guard against input saturation), and a *spatial heatmap* of final error (the structure of success and failure). Because the plant is stochastic and brains vary, robustness is examined separately by re-running across seeds and by perturbing the controller's own parameters.

6 Results and performance analysis

Aggregate performance. On the official grid (Fig. 3) the final distance is 32.3 ± 33.4 cm, the shoulder error $19.3 \pm 23.1°$, and the elbow error $53.1 \pm 63.0°$, with time-to-threshold 401 ± 211 steps and near-constant control effort (617 ± 23). In every metric the standard deviation is comparable to the mean: the average hides a sharply *bimodal* outcome.

A directionally-structured failure. Splitting the grid by workspace half makes the bimodality explicit:

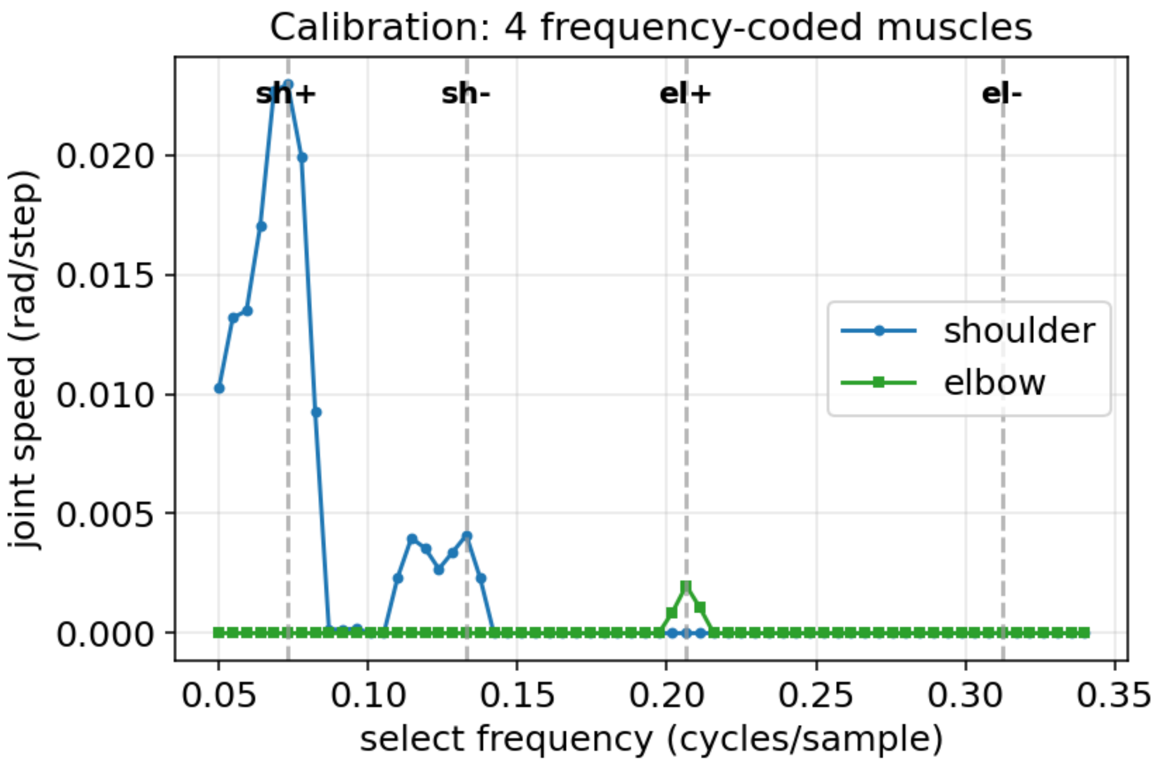

Figure 2: Offline calibration. Sweeping the select frequency and measuring the steady joint angular velocity reveals four band-pass muscle peaks — two per joint — giving the brain-invariant (frequency → joint, direction) map. The peaks at higher frequency are weaker, the first signature of the low-pass brain.

half	final dist.	|shoulder err|	|elbow err|
upper ($y > 0$)	5.7 cm	7°	7°
lower ($y < 0$)	61.6 cm	47°	101°

The upper half is reached to a few centimetres; the lower half misses by ~ 60 cm, and the residual is overwhelmingly elbow error. The cause is the low-pass brain (§2): the two halves recruit different members of each antagonist pair. Measured joint authority falls monotonically with frequency — shoulder+ (0.073) gives 9.7 rad of travel, shoulder− (0.133) 2.0, elbow+ (0.207) 2.0, elbow− (0.312) only 0.9. The upper half uses the strong, low-frequency muscles; the lower half needs the weak, high-frequency ones, above all the near-dead elbow−, which cannot bend the elbow the required way. That the control effort is flat across radius (Fig. 3f) confirms this is an *actuation* limit: the failing targets expend effort without converging, so more stimulation would not help.

Generalisation across brains. Calibration is performed once but every trial is a different random brain. Across five seeds on a common target set the controller reaches a median of 10.2 cm with 93% of trials within 15 cm: the brain-invariant structure transfers, and the loop absorbs each brain's gain.

Value of feedback (V1 vs. V2). Sharing the entire servo and differing only in the joint estimate, V1 (hand feedback) reaches a median 10.2 cm while V2 (neural observer, no hand feedback) reaches 33.7 cm with 0% within 15 cm. The observer runs correctly but drifts on the ∼10× noisier single-sample neural; the ∼24 cm gap quantifies the worth of direct hand sensing under this plant's noise.

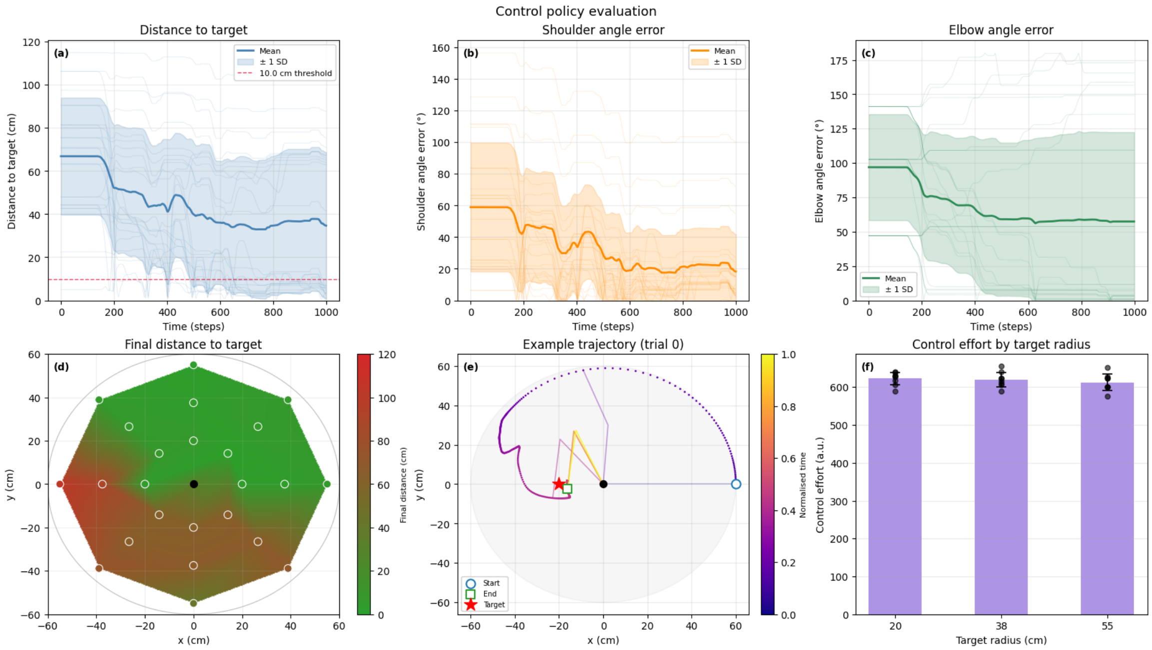

Figure 3: Standardised evaluation on the polar target grid (seed 0, $T = 1000$). (a) Distance-to-target: a $\sim$150-step power ramp, then a plateau with a wide band — the signature of a bimodal outcome. (b,c) Shoulder vs. elbow angle error: the shoulder error falls but the *elbow* error stays high, localising the bottleneck. (d) Final-distance heatmap: the failure is sharply *directional* — a reachable upper/right region (green, ≤ 12 cm) and a red lower/far-left shortfall (53–103 cm). (e) Example trajectory: the hand sweeps over the top and curls in. (f) Effort is flat across radius, so the failing targets spend effort without converging — an actuation limit, not under-driving.

Parameter sensitivity, and a cautionary result. On the calibration brain alone, raising the elbow gain improves accuracy, suggesting an easy win. *Cross-seed validation overturns this*: reliability (within 15 cm) *falls* from 90% at the default boost to 65% at a larger one, because over-driving the elbow on brains where it is adequate induces overshoot. The single-seed gain was over-fitting. This is itself a methodological finding — on a stochastic, per-brain-varying plant, single-brain tuning is misleading and parameters must be chosen by cross-seed reliability. The deadzone tolerance, by contrast, is nearly inert, showing the loop is not delicate.

7 Strengths, limitations and alternatives

Strengths. The controller is simple, requires no online model inversion, generalises across brains, and exploits rather than fights the decoder's structure (frequency selection plus free power). Its failure is interpretable and localised to a single physical cause.

Limitations. (i) *Elbow under-actuation* dominates: the higher-frequency muscles are low-pass-attenuated, so interior/lower targets are reach-limited (the directional failure above). (ii) *Shared input budget*: two simultaneous tones split the $[0, 1]$ amplitude, starving the weak elbow most. (iii) *Overshoot*: the head's slow power ramp and fast exit make the smallest effective stimulation larger than the final correction, so the hand hunts near the target rather than settling. (iv) *Dependence on hand sensing*: without it, V2 drifts (10$\to$34 cm). (v) *Stochastic stalls* from measurement noise, recovered by feedback at a cost in time.

Comparison with alternatives. The rejected model-based strategies (LQG, Fourier inversion, MPC) each fail a hard requirement — closing the loop on the hand and/or crossing the nonlinearity without a full model — whereas the select-tone servo does both by using the head's structure. Within the group, the sharpest contrast is with a *time-domain primitive finite-state-machine* controller: it rests on the *same* four rhythmic primitives but drives them as discrete, FSM-supervised bursts rather than continuous amplitude-proportional tones. Strikingly, the two structurally different controllers converge on the *same* precision wall — the minimum effective stimulation exceeds the fine correction. This is strong evidence that the binding limit is a property of the *plant* (the power ramp/exit asymmetry and stiction granularity), not of either control law, and is the most important cross-approach lesson.

8 Future improvements and conclusion

Three improvements follow directly from the analysis. First, *IK-branch selection by muscle strength*: most targets admit two arm configurations (elbow-up/down); the controller currently takes the one nearest the home pose, which for the lower half forces the weak elbow$-$. Choosing the branch that recruits the strong, low-frequency muscles would, in software alone, turn much of the red lower half green — and the example trajectory (Fig. 3e), which reaches a lower target by sweeping over the top with the strong muscle, shows this already works when geometry permits. Second, a *per-trial elbow self-calibration* rather than a fixed gain that over-fits one brain. Third, *sub-primitive actuation* — graded micro-bursts or amplitude annealing on approach — to beat the overshoot wall that no high-level controller removes.

In summary, exploiting the decoder as a frequency-addressed muscle bank with free power yields a controller that is simple, generalises across brains, and localises its failure to one physical bottleneck: the weak, high-frequency elbow muscles of a low-pass brain. The work also yields two transferable lessons — that direct hand sensing is worth $\sim$24 cm here, and that on a stochastic, varying plant, rigorous evaluation must be cross-seed, because best-case tuning misleads.